

Satvik Annadanam

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PERSONAL SUMMARY

Aerospace engineer with a Cranfield MSc in Astronautics & Space Engineering and a mechanical engineering background, grounded in aerodynamics, compressible flow and fluid dynamics. Experienced in airframe structural optimisation (CATIA V5, FEA), propulsion trade studies, and simulation-driven design across FEA, multiphysics and RF environments. Comfortable working at the concept and systems level, making engineering trade-offs under incomplete data, and turning analysis into design decisions. Genuine interest in unmanned and non-traditional aircraft and in tight airframe-propulsion integration, with hands-on UAV and eVTOL platform exposure through antenna and propulsion projects. Strong in V&V methodology and cross-disciplinary collaboration.

KEY ACHIEVEMENT

- Won “Highest to reach” in the rocket competition in which collaborated in a team of 5 to build a rocket from scratch within a 2-hour time limit (Cranfield University 2024).
- Won the “Low Light Photography” competition as a result of outstanding image capture of a firework display (SASTRA University 2023).

EDUCATION

MSc in Astronautics and Space Engineering, Cranfield University, Cranfield, UK **Oct 2024 - Sep 2025**

- **Modules:** Space Systems Engineering, Space Propulsion, Astrodynamics & Mission Analysis, Payload Engineering, Computational Mathematics for Trajectory Optimisation, Advanced Composite Structures.
- **Group Project:** Reusability of Lunar Landers - Engineering a sustainable propulsion system, optimising tank design, propellant selection, and system integration to support long-term lunar operations.
- **Individual Project:** Design of Compact Phased Array for UAV Platform - Designed compact and low-profile antenna arrays suitable for various system integration on UAVs, HAPS, and LEO.

BTech. Mechanical Engineering, SASTRA University, Thanjavur, India **Sep 2020 - Jul 2024**

- **Modules:** Compressible Flow, CAD/CAM, Fluid Dynamics, Heat Transfer System Design, Industry 4.0 and IIoT.
- **Major Project:** Composite Aircraft Structural Optimisation - Applied computational mechanics and finite element analysis to improve structural efficiency through advanced material selection and structural reinforcement techniques.

INTERNSHIP EXPERIENCE

Thermal Engineer Intern - Nitrexo **Sep 2025 - Feb 2026**

Nitrexo is an engineering firm focused on simulation-driven design and AI-assisted analysis tools for aerospace and advanced systems.

Accomplishments:

- Conducted geometry analysis and simplification of spacecraft subsystems using ANSYS, MSC Apex, and SolidWorks to prepare models for thermal simulations.
- Designed and optimised algorithmic workflows to automate geometry preparation and analysis steps, enabling scalability across multiple spacecraft configurations.
- Authored a Verification & Validation (V&V) Plan, executed validation tests, and compiled a comprehensive V&V Report ensuring software accuracy and reliability.
- Collaborated with the software development team to integrate automation features, bridging mechanical engineering and software workflows.
- Delivered a live demonstration comparing manual vs. automated workflows, quantifying improvements in analysis speed and consistency.

Internship Student - Calderys **Jun 2023 - Jul 2023**

Calderys is a global leader in thermal protection, providing refractory products and advanced solutions for high-temperature industries. It specialises in enhancing steel casting, metallurgical fluxes, and moulding processes.

Accomplishments:

- Performed mechanical assembly and gear replacement tasks: identified gear specifications, ensured proper meshing, and aligned components for efficient machine operation.
- Maintained bearing mechanisms by disassembling, inspecting for wear, re-lubricating, and replacing damaged components; applied welding adjustments to minimise machine downtime.
- Monitored SCADA systems to regulate temperature and pressure in high-heat processes (300–400°C range), overseeing valve operations and processing of 2 tonnes of material.
- Ensured compliance with industrial safety protocols across heat, height, vehicle, cutting, and lifting operations, reinforcing adherence to workplace safety standards.

RESEARCH EXPERIENCE

GEO Orbit Perturbations for Radar Interferometry (HydroGEO)

Apr 2026 - Sep 2026

Research Assistant - Cranfield University

Research project linked to the ESA HydroGEO mission concept, investigating long-term orbit control of a geostationary radar satellite to enable bistatic interferometric synthetic aperture radar (InSAR).

Accomplishments:

- Built working understanding of the four dominant GEO perturbations: luni-solar gravity, Earth triaxiality (J22), solar radiation pressure, and J2; and their impact on long-term secular drift versus short-period oscillations.
- Preparing GMAT simulations to propagate a representative GEO orbit with all four perturbations over multi-week to multi-month timescales, supporting baseline-orbit control studies.

Design of Compact Phased Array for UAV and HAPs Platform

Apr 2025 - Sep 2025

Individual Project - Cranfield University

Designed a multi-band shared-aperture antenna for an eVTOL platform (Vertical Aerospace VX4) using ANSYS HFSS and MATLAB/Simulink, supervised externally by Honeywell. Targeted simultaneous Ku-band (OneWeb) and L-band (Iridium) satellite connectivity for high-altitude aerial systems.

Accomplishments:

- Developed a compact shared-aperture architecture minimising mutual coupling, substrate height, and inter-element spacing inspired by Honeywell SMALL UAV antenna systems.
- Modelled antenna performance across flight dynamics using HFSS and MATLAB/Simulink: return loss, radiation patterns, and in-flight link reliability.
- Built a MATLAB algorithm for dynamic antenna and link adjustment, compensating for environmental effects such as Ku-band rain fade to maintain robust connectivity.

Propulsion System Team - Abomination of Single-Use Lunar Landers

Group Design Project - Cranfield University

Oct 2024 - Apr 2025

Redesigned the propulsion system using SolidWorks. Conducted trade studies across chemical bipropellant, monopropellant and electric propulsion. Collaborated with interdisciplinary teams (structures, avionics, architecture, power, and end-of-life systems) to ensure system integration.

Accomplishments:

- Conducted a thorough review of IIT Argonaut field research and prepared a comprehensive general requirements document for mission feasibility.
- Owned tank geometry, propellant selection, valve mechanisms, and nozzle architecture; proposed modifications supporting end-of-life recyclability (15 kg mass penalty accepted for reduced orbital debris risk).
- Conducted trade studies across chemical bipropellant, monopropellant, Hall Effect Thrusters (HET), and Gridded Ion Engines (GIE), prioritising low-contamination solutions for long-term orbital sustainability.

Team Leader – Composite Aircraft Structural Optimisation

Jan 2024 - Jun 2024

Major Project - Indian Institute of Technology Kanpur

Redesigned an aircraft wing structure using CATIA to improve load-bearing capacity and structural stability under high-stress conditions. Applied advanced composite materials to critical structural components and validated performance gains

Accomplishments:

- Modelled a tapered aircraft wing in CATIA V5 and SolidWorks, building the geometry from NACA six-series laminar-flow airfoils (64A215 root, 64A210 tip), a 5.5 m semi-span and a two-spar layout across 15, six and four - rib configurations.
- Set up the parametric design matrix: five candidate materials (aluminium 2024-T3 plus four epoxy composites) against three rib counts - to support material and structural selection on a stiffness-versus-weight basis.
- Co-authored the resulting static-structural and modal FEA study (ANSYS Workbench): aluminium showed lowest deflection while epoxy-carbon UD gave the best stiffness-to-weight, with modal analysis identifying natural frequencies and mode shapes for each configuration.
- Proposed next-generation composite reinforcement methodologies to enhance structural performance.

Team Leader – Robotic Manipulator

Jan 2023 - May 2023

Minor Project - SASTRA University

Accomplishments:

- Designed and built a 7-DOF robotic arm in SolidWorks with 3D-printed structural components, integrating Arduino/ROS control and Alexa voice operation.
- Built a real-time RViz digital twin for kinematic validation and trajectory verification, achieving approximately 15% better operational efficiency than an Arduino-only baseline.

SKILLS, INTERESTS AND EXTRACURRICULAR ACTIVITIES

- Languages: English (fluent), Hindi (fluent), Telugu (fluent), Marathi (fluent), German (basic).
- Software & Tools: SolidWorks, CATIA V5, PTC Creo, ANSYS (Workbench, HFSS, Apex), COMSOL, MATLAB/Simulink, Python, Microsoft Office. Learning: GMAT, Fusion 360, CFD. Basic: LabVIEW, C++, OpenRocket STK.
- Memberships: CRANSEDS (Propulsion Team), Robotics Club, Photography Society, Astronomical Society.
- Interests: Passionate about football (CR7), Tennis (Wimbledon & Nadal on clay), Formula 1 (Charles Leclerc, Scuderia Ferrari), enjoy long-exposure astrophotography, capturing celestial bodies like Jupiter and Mars. Enthusiastic about anime (seinen and shōnen), with favourites such as Attack on Titan, Berserk, JOJO.
- Certifications: Python, low-light photography, MATLAB/Simulink.